

Stationarity for the AR(1) Process

We have $X_t = \phi X_{t-1} + Z_t$. We can write this as

$$X_t = \sum_{j=0}^{\infty} \phi^j Z_{t-j}$$

Clearly $E(X_t) = 0$

$$\text{Var}(X_t) = \text{Var}\left(\sum_{j=0}^{\infty} \phi^j Z_{t-j}\right)$$

$$= \sum_{j=0}^{\infty} \phi^{2j} \text{Var}(Z_{t-j})$$

independence
of $\{Z_t\}$
terms

$$= \sigma_Z^2 \sum_{j=0}^{\infty} \phi^{2j}$$

$$\begin{aligned} \text{Cov}(X_t, X_{t+h}) &= E(X_t X_{t+h}) \\ &= E\left[\left(\sum_{j=0}^{\infty} \phi^j Z_{t-j}\right) \left(\sum_{k=0}^{\infty} \phi^k Z_{t+h-k}\right)\right] \\ &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \phi^{j+k} E(Z_{t-j} Z_{t+h-k}) \end{aligned}$$

Because $\{Z_t\}$ is a white noise process

$$E(Z_{t-j} Z_{t+h-k}) = 0 \text{ except where}$$

$$-j = h-k \Rightarrow k = j+h.$$

$$\begin{aligned}
 \therefore \gamma(h) &= \text{Cov}(X_t, X_{t+h}) \\
 &= \sum_{j=0}^{\infty} \phi^{2j+h} \mathbb{E}(Z_{t-j}^2) \\
 &= \sigma_z^2 \phi^h \sum_{j=0}^{\infty} \phi^{2j}
 \end{aligned}$$

$$\gamma(-h) = \gamma(h)$$

Consider $|\phi| < 1$

If $|\phi| < 1$ then $\sum_{j=0}^{\infty} (\phi^2)^j$ converges

(sum of a geometric series $a = 1$
 $r = \phi^2$)

$$\therefore \sum_{j=0}^{\infty} (\phi^2)^j = \frac{1}{1 - \phi^2}$$

$$\therefore \gamma(h) = \frac{\sigma_z^2 \phi^h}{1 - \phi^2}$$

Consider $|\phi| > 1$

$$\text{Var}(X_t) = \sigma_z^2 \sum_{j=0}^{\infty} (\phi^2)^j \rightarrow \infty.$$

$\therefore$ Not stationary if $|\phi| > 1$.

Consider $|\phi| = 1$

$$X_t = \pm X_{t-1} + Z_t$$

$$\Rightarrow X_t^2 = X_{t-1}^2 - 2X_{t-1}Z_t + Z_t^2$$

Take expectations of both sides

$$\begin{aligned} \mathbb{E}(X_t^2) &= \mathbb{E}(X_{t-1}^2) \pm 2 \mathbb{E}(X_{t-1} \overset{\circ}{Z_t}) + \sigma_Z^2 \\ &\quad \text{causality} \\ &= \mathbb{E}(X_{t-1}^2) + \sigma_Z^2 \end{aligned}$$

$$\Rightarrow \text{Var}(X_t) = \text{Var}(X_{t-1}) + \sigma_Z^2$$

$\therefore \text{Var}(X_t)$ is increasing as t increases.

$\therefore$ Not stationary if $|\phi| = 1$.